01 / PRODUCT AND PROBLEM

From a reported symptom to a source-linked fix plan.

BugLens is a repository-first review workspace and reusable agent skill for developers and maintainers. It combines a public GitHub repository with optional text, logs, source attachments or screenshots to produce investigation leads and a practical coding-agent handoff.

The problem

A screenshot or error message rarely identifies the exact source of a bug. Reviewing a repository and turning incomplete evidence into an actionable report takes effort. BugLens organizes that evidence; it does not promise to diagnose every defect or automatically repair code.

Three implemented capabilities

1. Review the repository. Commit-pinned source import, deterministic checks, Gemma 4 review and npm advisory matching. Gemma source review is enabled by default in the web app; users control external checks and consent.

2. Add the missing context. Investigate a reported problem using supported text/code files and optional screenshot evidence.

3. Leave with a plan. Source-linked findings, Markdown report, remediation documentation and combined or per-finding fix prompts.

The distinctive mechanism

Source findings carry path/line evidence. Model source references must match supplied excerpts exactly. Screenshot observations, reporter statements and suspected causes remain distinct. Every finding stays marked as needing verification; coverage records what was not checked.

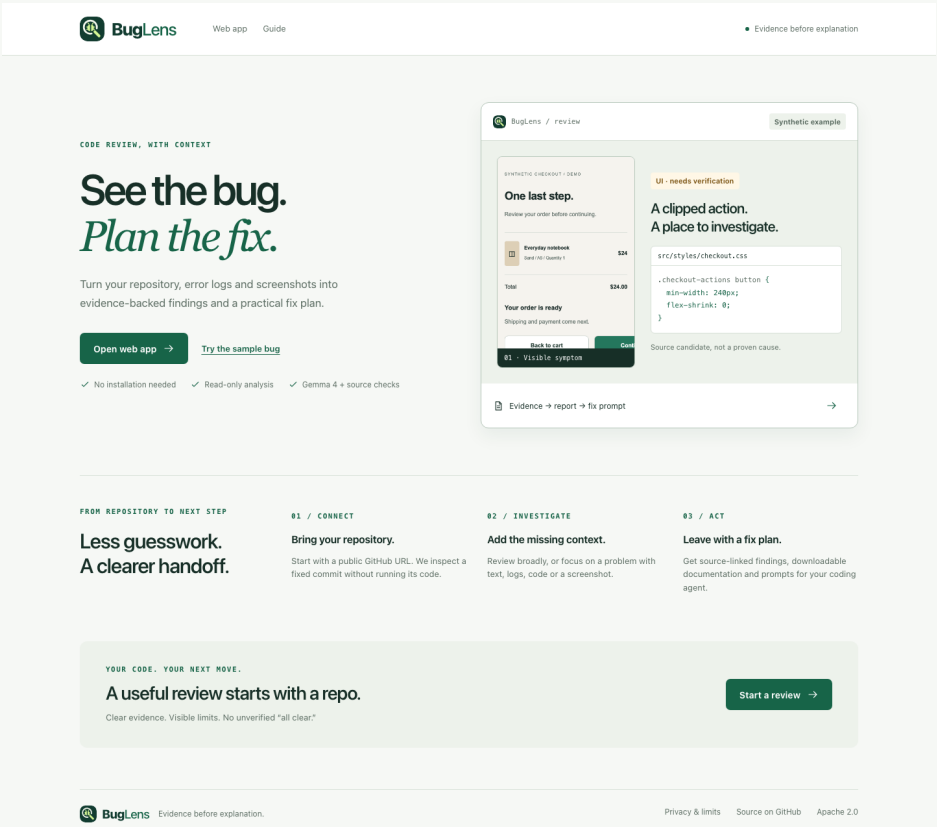
Official challenge alignment: the problem statement and submission requirements were not supplied, so alignment and eligibility cannot be assessed. Time saved, accuracy and competitive advantage have not been measured.

Repository: github.com/yatharthvishnoi08-max/buglens
Live workspace: buglens-rust.vercel.app/demo

02 / USER JOURNEY

Repository first. Evidence in context.

Connect a public repository. **Choose** broad review or a reported problem. **Add** optional evidence. **Select** external checks and consent. **Inspect** findings and coverage. **Download** the report, documentation or fix prompt.



Actual local browser capture of the redesigned landing page. Its product preview is explicitly synthetic. The redesigned UI and refreshed logo were published in release a6605e5. The web app needs no installation; the agent-skill ZIP is a separate developer tool.

What users receive

buglens-report.md: scope, findings and limitations.
buglens-remediation.md: problem register and acceptance checklist.
buglens-fix-prompt.txt: an investigation prompt for an authorized coding agent.

The prompt is not a shell command, an automatic patch or proof of root cause. Download artifacts before refreshing: results are kept only in browser memory.

03 / ARCHITECTURE AND TRUST

A bounded review pipeline.

What runs

Browser: Next.js 16 / React 19 / TypeScript views manage inputs and render findings. Presentation components are separated from request/state containers.

Node API and reusable skill: both call the shared repository-review pipeline. The API accepts multipart options and an optional image, then streams progress and a result or explicit error as NDJSON.

Processing: resolve the GitHub default branch to a commit; parse a bounded archive in memory; exclude unsafe/sensitive-looking paths; run static patterns and requested external checks; validate source references; compile artifacts.

External services and storage

GitHub supplies public source. **Google** receives selected source/problem inputs or screenshot context when AI is enabled. The only permitted model is gemma-4-26b-a4b-it; Gemini is the API transport. **OSV** receives package names and versions for requested advisory checks.

There is no database, login or saved review history. Source archives and outputs are processed in memory. Provider/hosting handling is separate. Credential redaction is partial: users must not submit secrets.

Deliberate boundaries

No repository code, installation scripts, build, tests or running browser are executed. No automatic issue creation, push, deployment or repair occurs. Public repositories/default branch only; PDFs, Word, audio and video are not parsed.

Limits: 1,000 supported source files / 8 MB total; up to 16 AI excerpts / 45,000 source characters; up to 200 exact npm versions. Text inputs are bounded; production screenshots are PNG/JPEG/WebP up to 4 MB.

Failures remain visible

Requested AI failure returns an error, not another model or hidden fixture output. OSV failure can return a partial report with an unavailable coverage status. Zero findings is not an all-clear.

Implementation references: src/lib/review/{github,checks,model,report,run}.ts; src/model/gemma-client.ts; src/app/api/review-repository/route.ts. See docs/architecture.md for contracts and data boundaries.

04 / VERIFICATION AND READINESS

Show the evidence. Keep the limits visible.

Observed on 26 October 2026: macOS, Node 20.18.0, Next.js 16.4.3, Chromium. Release a6605e5 was checked locally and its public web-app/download flow was checked in desktop and mobile browsers. These are not hosted CI certification.

Verification summary

PASS - TypeScript check and production build.
PASS - 94 unit/contract tests across 8 files.
PASS - 4 deterministic synthetic evaluation cases.
PASS - 7 browser tests; 1 duplicate matrix invocation intentionally skipped.
PASS - Route/overflow checks at 375, 768, 1024 and 1440px.
PASS - 2 public-site browser checks: desktop and mobile entry/download flow.

Browser checks cover text/file input, artifact downloads, keyboard entry/focus, memory reset, recoverable error and partial results. Repository browser responses are controlled data from the real static pipeline. The screenshot sample uses pre-authored model output. Neither proves live source-AI success.

Open evidence gaps

Newer live Gemma source review remains unverified after historical malformed-output, HTTP 500 and HTTP 429 failures. Earlier screenshot-only success is not equivalent. No full accessibility audit, cross-browser test, fresh package installation, runtime target-repository audit or measured model-accuracy benchmark is claimed.

Historical dependency audit reported three development-tool advisories. This UI/documentation work did not change the test toolchain. See docs/TEST_REPORT.md for commands, evidence and failed checks that were corrected and retested.

Team, publication and next steps

Team: names, roles and contributions were not supplied. Repository ownership is not proof of authorship. **Submission:** not yet filed; official requirements still need review.

The repository-first UI, refreshed logo and web-app/download guidance are published on GitHub and Vercel. Next: confirm team/challenge details, validate live source-AI when quota permits, record the evidence-backed 90-second demo and submit the required files.

Guide: docs/DEMO_GUIDE.md | Evidence: docs/TEST_REPORT.md
Repository: github.com/yatharthvishnoi08-max/buglens
Demo: buglens-rust.vercel.app/demo